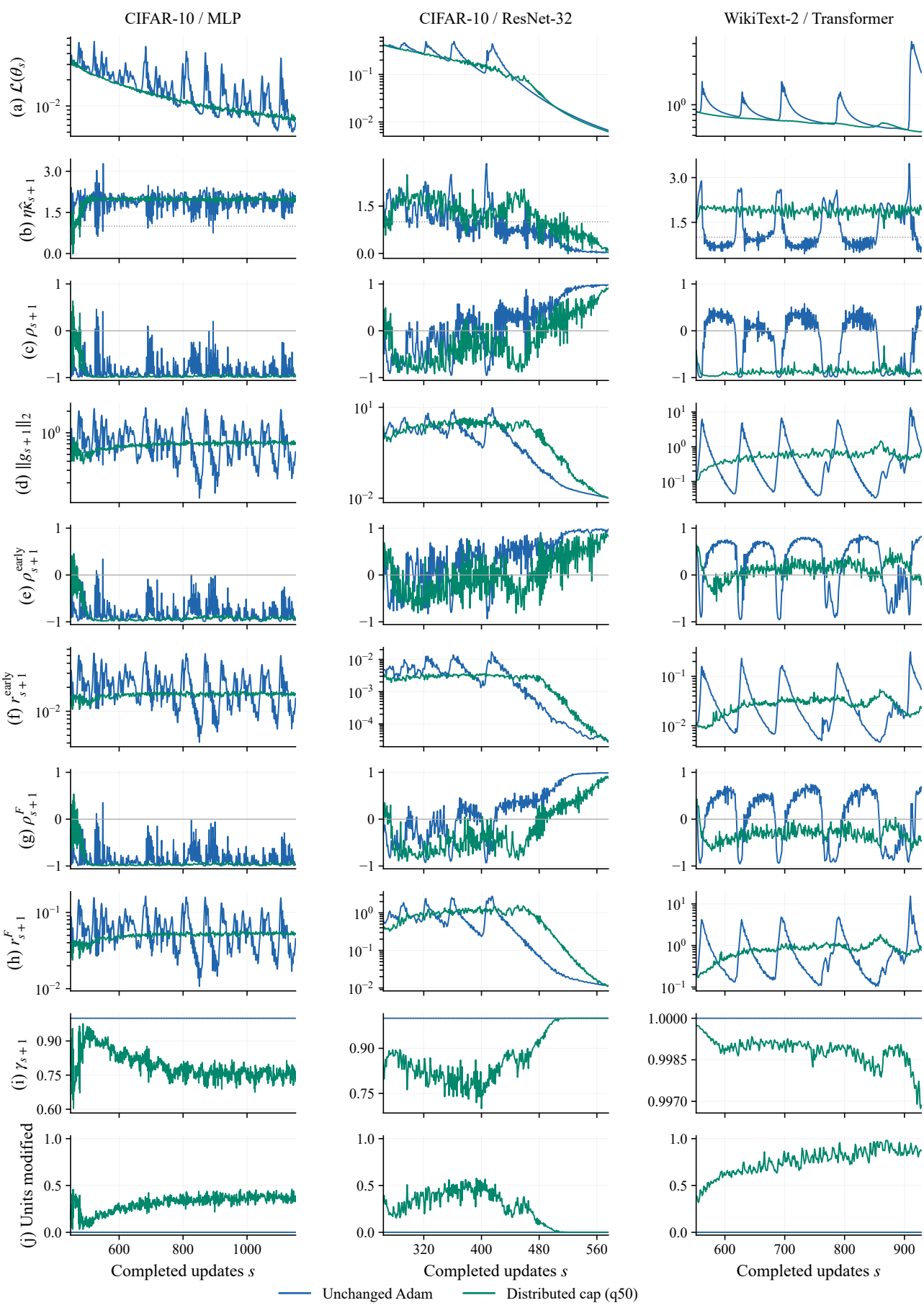

CIFAR-10 / MLP
CIFAR-10 / ResNet-32
WikiText-2 / Transformer
(a) $\mathcal{L}(\theta_s)$
(b) $\eta\hat{\kappa}_{s+1}$
(c) ρ_{s+1}
(d) $\|g_{s+1}\|_2$
(e) $\rho^{\text{early}}_{s+1}$
(f) r^{early}_{s+1}
(g) ρ^{F}_{s+1}
(h) r^{F}_{s+1}
(i) γ_{s+1}
(j) Units modified
Completed updates s
Completed updates s
Completed updates s
Unchanged Adam
Distributed cap (q50)